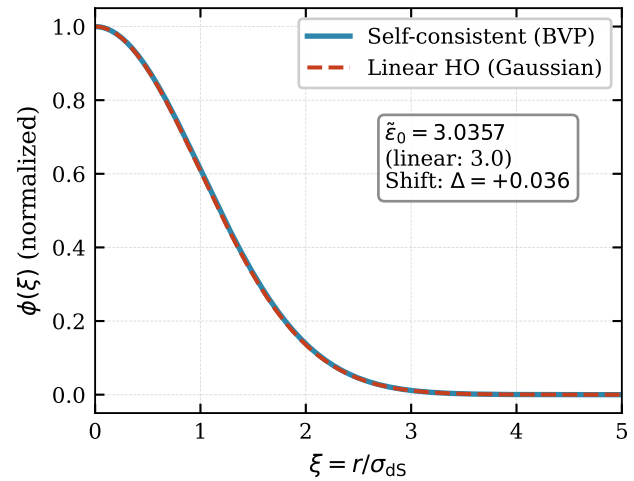
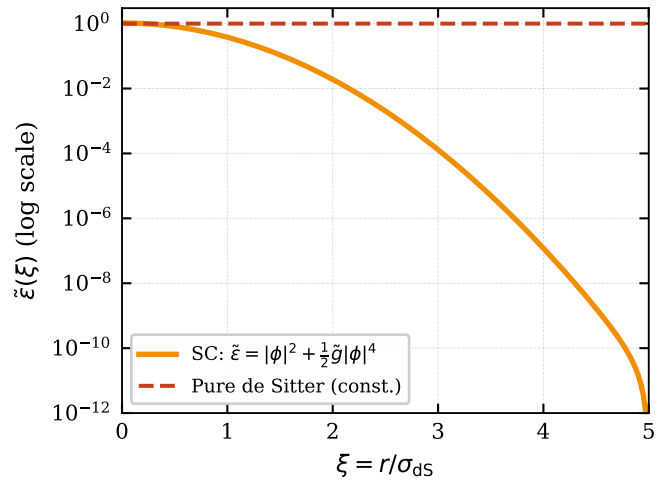


Self-consistent Einstein--Gross--Pitaevskii system: ground-state condensate in the de Sitter harmonic trap

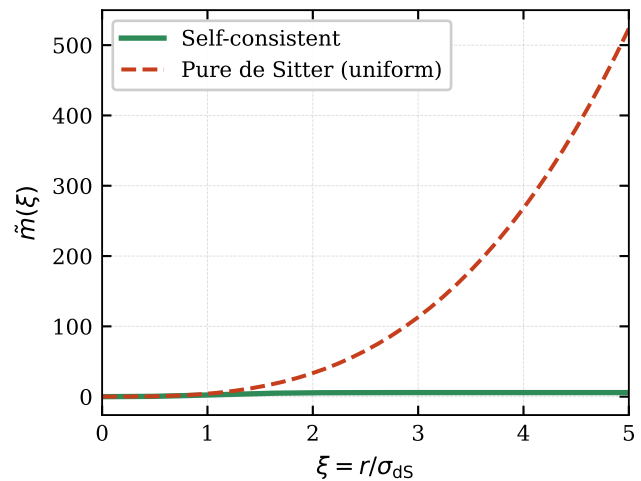
Condensate Profile (Ground State)



Energy Density: Localized vs. Uniform



Enclosed Mass-Energy



Self-Interaction Effect on Ground State

